

Year 7 • Chapter 1 • Integers: answers (teacher copy)

1.04 Adding integers (booklet p. 3)

Teacher notes

Key misconception. Treating $-2 + 5$ as “ $-(2 + 5)$ ” (the symmetric error), and carrying “two negatives make a positive” over from multiplication into addition.

Why? (pair 1) We moved right. Why?

We added a positive number (+3), and adding a positive moves right.

Why? (pair 2) $2 - 5$ also moves 5 left from 2. What does that tell you about $2 + (-5)$?

$2 + (-5)$ is the same as $2 - 5$. Adding a negative is the same as subtracting.

Why? (pair 3) Tom says $-4 + (-3) = 7$. What went wrong? Tom used “two negatives make a positive”. That rule is about multiplying, not adding. Adding a negative moves left, so $-4 + (-3) = -7$.

Check (hinge). Work out $-2 + 5$ Answer: B .
A (-7): added the sizes and kept the minus sign (“symmetric” error). Re-teach with Example 1 on the number line.
B (3): correct.
C (-3): moved the wrong way, or did $5 - 2$ and kept the minus. Act it out: start at -2, walk 5 steps right.
D (7): ignored the negative sign. Check they read -2 as a place left of zero (Words box).

Next. Most students B: go on to page 2. Students with A, C or D: a short re-teach at the board with Examples 1 and 2 on the number line (walk it out), then Your turn 1 and 2 again before page 2. Students who finish Mixed practice: Go further.

Think first. Take answers without judging. Show a thermometer: start at -3, count up 5. Link to Example 1: the same move on a sideways number line.

Worked answers

Do now 1 $-7 \square -2$ Write $<$ or $>$.
 $-7 < -2$

Do now 2 Order smallest to largest: 4, -6, 0, -1
 $-6, -1, 0, 4$

Do now 3 What is the opposite of -9?
9

Do now 4 Start at -4. Move 6 right. Where do you land?
2

Think first It is -3°C at 6 am. By noon it is 5°C warmer. What is the temperature at noon?
 2°C

Example 1 $-5 + 3$
start at -5, move 3 right $\cdot = -2$

Your turn 1 $-6 + 4$
start at -6, move 4 right $\cdot = -2$

Example 2 $2 + (-5)$
start at 2, move 5 left $\cdot = -3$

Your turn 2 $3 + (-7)$
start at 3, move 7 left $\cdot = -4$

Example 3 $-4 + (-3)$
 $= -4 - 3 \cdot = -7$

Your turn 3 $-2 + (-6)$
 $= -2 - 6 \cdot = -8$

Pattern 1 $-3 + 3$
0

Pattern 2 $-3 + 2$
-1

Pattern 3 $-3 + 1$
-2

Pattern 4 $-3 + 0$
-3

Pattern 5 $-3 + (-1)$
-4

Pattern 6 $-3 + (-2)$
-5

Pattern 7 $-3 + (-3)$
-6

Pattern 8 $-30 + (-30)$
-60

Notice What happens to the answer each time the number added goes down by 1?
The answer goes down by 1 too. Adding a negative carries on the pattern: it makes the answer smaller.

Mixed 1 $-8 + 5$
-3

Mixed 2 Which is larger: -8 or 5?
5

Mixed 3 $6 + (-9)$
-3

Mixed 4 How far apart are -8 and 5?
13

Mixed 5 $-7 + (-2)$
-9

Mixed 6 What is the opposite of -12?
12

Mixed 7 $-15 + 15$
0

Mixed 8 $-6 + \square = -1$
 $\square = 5$

Mixed 9 A diver is at -12 m. She rises 5 m. Where is she now?
 $-12 + 5 = -7$ m

Mixed 10 Order smallest to largest: -3 + 4, -3, -3 + (-4)
 $-3 + (-4), -3, -3 + 4$ (that is $-7, -3, 1$)

Go further 1 Always, sometimes or never true? “Adding a negative number makes the answer smaller.” Explain.
Always. Adding a negative always moves left, so the answer is always smaller than the starting number.

Go further 2 Find two integers that add to -3 . Then find two more, where one of them is positive.

e.g. $-1 + (-2) = -3$; **e.g.** $2 + (-5) = -3$

Go further 3 Start at -4 . Add 2, 6 and -5 in any order. Where do you finish? Does the order matter?

$-4 + 2 + 6 + (-5) = -1$. **The order does not matter.**

Exit 1 $-7 + 3$

-4

Exit 2 $4 + (-9)$

-5

Exit 3 It is -6°C . It warms up 10°C . What is the temperature now?

$-6 + 10 = 4^{\circ}\text{C}$